

# Paloma Sodhi

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## SUMMARY

Research Scientist with ML and NLP expertise working on dialog agents that converse and act. Specializing in large language models, reinforcement learning from human/AI feedback, and LLM agents. Experienced in leading independent research projects, training large-scale models, and deploying them in production.

## WORK EXPERIENCE

### ML/NLP Research Scientist

May 2022 – Present

*ASAPP Research*

*New York City, NY*

- Building ML and NLP methods for customer service chatbots that both converse with customers and take actions.
- Trained LLaMA3-8b, 70b models with AI feedback for dialog agents, tool use, and coding, showing smaller models can outperform GPT-4o when using privileged information.<sup>[1]</sup>
- Led development of new LLM agent architectures that decompose complex tasks into modular policies. Open-sourced models achieving top performance on WebArena leaderboard<sup>[2]</sup>. Deployed in production.<sup>[7]</sup>
- Investigated PPO vs offline RL<sup>[3,4]</sup> and preference learning methods like DPO, KTO for conversational agents.

### Visiting Researcher AI

Mar 2021 – Mar 2022

*Facebook AI Research*

*Pittsburgh, PA*

- Developed efficient inverse reinforcement learning methods to learn reward models for large-scale factor graphs.<sup>[6]</sup>
- Worked on an open-source differentiable optimizer library for training end-to-end models.<sup>[5]</sup>

### Graduate Researcher

Jan 2016 – Mar 2022

*Carnegie Mellon University*

*Pittsburgh, PA*

- Ph.D. on combining machine learning with graph optimizers in the loop, with applications to robot manipulation.

### AI Research Intern

Jun 2020 – Aug 2020

*Facebook AI Research*

*Remote*

- Built multimodal vision-tactile models for embodied AI applications like in-hand robot manipulation.

## EDUCATION

### Carnegie Mellon University

Pittsburgh, PA

*Ph.D. in Robotics, School of Computer Science, GPA: 4.0 / 4.0*

*Aug 2017 – Mar 2022*

### Carnegie Mellon University

Pittsburgh, PA

*M.S. in Robotics, School of Computer Science, GPA: 4.06 / 4.0*

*Aug 2015 – May 2017*

### Indian Institute of Technology Guwahati

Guwahati, India

*B.S. in Electrical Engineering*

*Aug. 2009 – May 2013*

## TECHNICAL SKILLS

**Languages:** Python, C/C++, JavaScript, HTML/CSS    **Libraries:** HF Transformers, PyTorch, JAX, Gym

## SELECTED PAPERS

- [1] S. Choudhury\* and **P. Sodhi\***. *Better than Your Teacher: LLM Agents that learn from Privileged AI Feedback International Conference on Learning Representations (ICLR) 2025*, in submission. \*equal contribution [\[paper\]](#) [\[code\]](#)
- [2] **P. Sodhi**, S. Branavan, Y. Artzi, R. McDonald. *SteP: Stacked LLM Policies for Web Actions. Conference on Language Modeling (COLM) 2024*. [\[paper\]](#) [\[code\]](#)
- [3] **P. Sodhi**, F. Wu, E. Elenberg, K. Weinberger, and R. McDonald. *On the Effectiveness of Offline RL for Dialogue Response Generation. International Conference on Machine Learning (ICML) 2023*. [\[paper\]](#) [\[code\]](#)
- [4] D. J. Min, **P. Sodhi**, R. Ramakrishnan. *Workflow-Guided Response Generation for Task-Oriented Dialogue. arXiv 2023*. [\[paper\]](#) [\[code\]](#)
- [5] L. Pineda, T. Fan, M. Monge, S. Venkataraman, **P. Sodhi**, et al. *Theseus: A library for differentiable nonlinear optimization. Neural Information Processing Systems (NeurIPS) 2022*. [\[paper\]](#) [\[code\]](#)
- [6] **P. Sodhi**, E. Dexheimer, M. Mukadam, S. Anderson, M. Kaess. *LEO: Learning Energy-based Models in Factor Graph Optimization. Conference on Robot Learning (CoRL) 2021*. [\[paper\]](#) [\[code\]](#)

## PATENTS

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- [7] **P. Sodhi**, R. McDonald, S. Branavan, V. Cirik *Automation of Tasks using Language Model Prompts*  
Patent No. 18/342,350, Filed: Jun 27, 2023, Pending.
- [8] S. Ray, K. Weinberger, S. Branavan, **P. Sodhi** *Automated Key-Value Extraction using Natural Language Intents*  
Patent No. US3456789C1, Filed: Dec 9, 2022, Pending.

## AWARDS

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- **FRAIM Research Scholarship**, Selected as 1 in 7 students from CMU School of Computer Science.  
By Facebook AI Research, USA · Mar 2022
- **University Presidential Fellowship** Towards excellence and impactful research by Ph.D. students.  
By Carnegie Mellon University, USA · Oct 2018
- **Best Conference Paper Finalist** Top 0.25% of accepted papers.  
By IEEE Conference on Intelligent Robots and Systems (IROS) · Sep 2021
- **Best Application Paper Finalist** Top 0.4% of accepted papers.  
By IEEE Conference on Intelligent Robots and Systems (IROS) · Sep 2017